网络安全新技术——

MTD

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Moving target defense

- Moving target defense(MTD) is a new way to make the network “dynamic” to attackers’ view.
- So far, researches have proposed about three kind of methods
  - Network mapping and reconnaissance protection
  - Service version and OS hiding
  - Random host or route mutation

Network mapping and reconnaissance protection

Algorithm 1 MTD against network reconnaissance

```
Require: Probabilities Pr_SA < Pr_A < Pr_PA < Pr_R < 1
  hash table action_buffer ← NULL
  while (new TCP packet p is received) do
    if (p is illegitimate traffic) then
      if (p.dest_port not in action_buffer) then
        r ← random real number ∈ [0, 1]
        store r in action_buffer
      else
        r ← as in action_buffer
      end if
      switch (r)
       case r < Pr_SA:
        respond with TCP SYN-ACK
       case r < Pr_A:
        respond with TCP ACK
       case r < Pr_PA:
        respond with TCP PUSH-ACK with random payload
       case r < Pr_R:
        respond with TCP RST packet
       default:
        drop silently
      end switch
    end if
  end while
```

Randomly respondence;
Protect from DoS attacks

Kampanakis, Panos, Harry Perros, and Tsegereda Beyene. "SDN-based solutions for Moving Target Defense network protection." *A World of Wireless, Mobile and Multimedia Networks (WoWMoM), 2014 IEEE 15th International Symposium on*. IEEE, 2014.

Service version and OS hiding

- Service version and OS hiding

Algorithm 2 MTD against OS fingerprinting

```
while (new TCP packet p destined to target is received) do
  if (p is illegitimate traffic) then
    if (p has TCP SYN set) then
      s ← random 32-bit number
      respond with TCP SYN-ACK and s as the seq#
    else
      generate random payload and respond
    end if
  end if
end while
```

When the TCP SYN-ACK seq has the information about OS or service version

Kampanakis, Panos, Harry Perros, and Tsegereda Beyene. "SDN-based solutions for Moving Target Defense network protection."
A World of Wireless, Mobile and Multimedia Networks (WoWMoM), 2014 IEEE 15th International Symposium on. IEEE, 2014.

Random Host Mutation

- Features:
  - 1. IP mutation is transparent to end host;
  - 2. IP mutation is performed with **high unpredictability and rate** to maximize the distortion of attacker’s knowledge and increase deterrence of attack planning.
- Principles: each host associate with an unused address range, in every mutation interval, picking up a virtual IP to associate with this host.
- contributions：
  - allocating unused ranges to hosts
  - mutate within allocated ranges
- Solving method - model as a constraint satisfaction problem, solved by the tool called Yices, a kind of SMT solvers

Random Host Mutation

- Process of communication

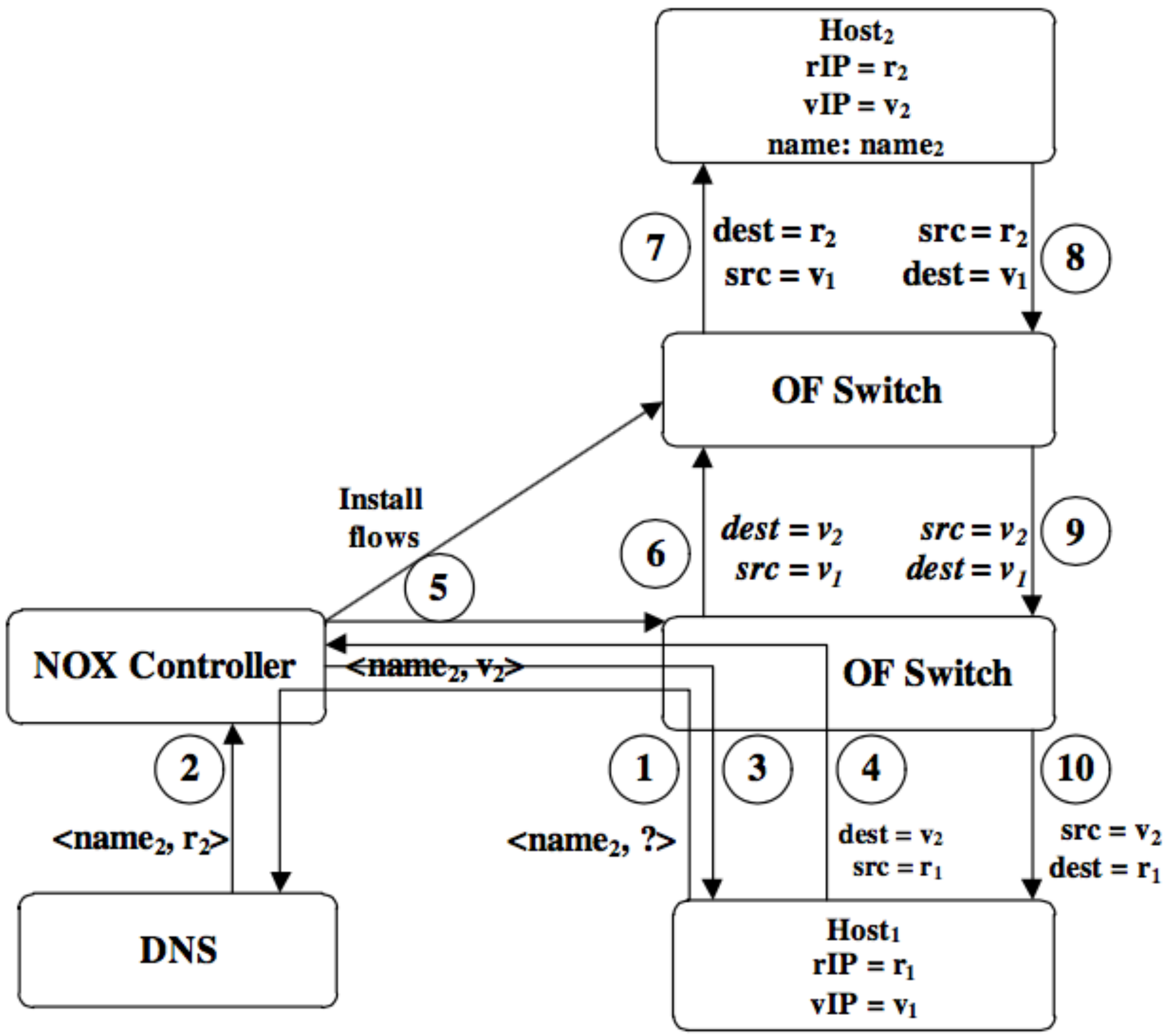
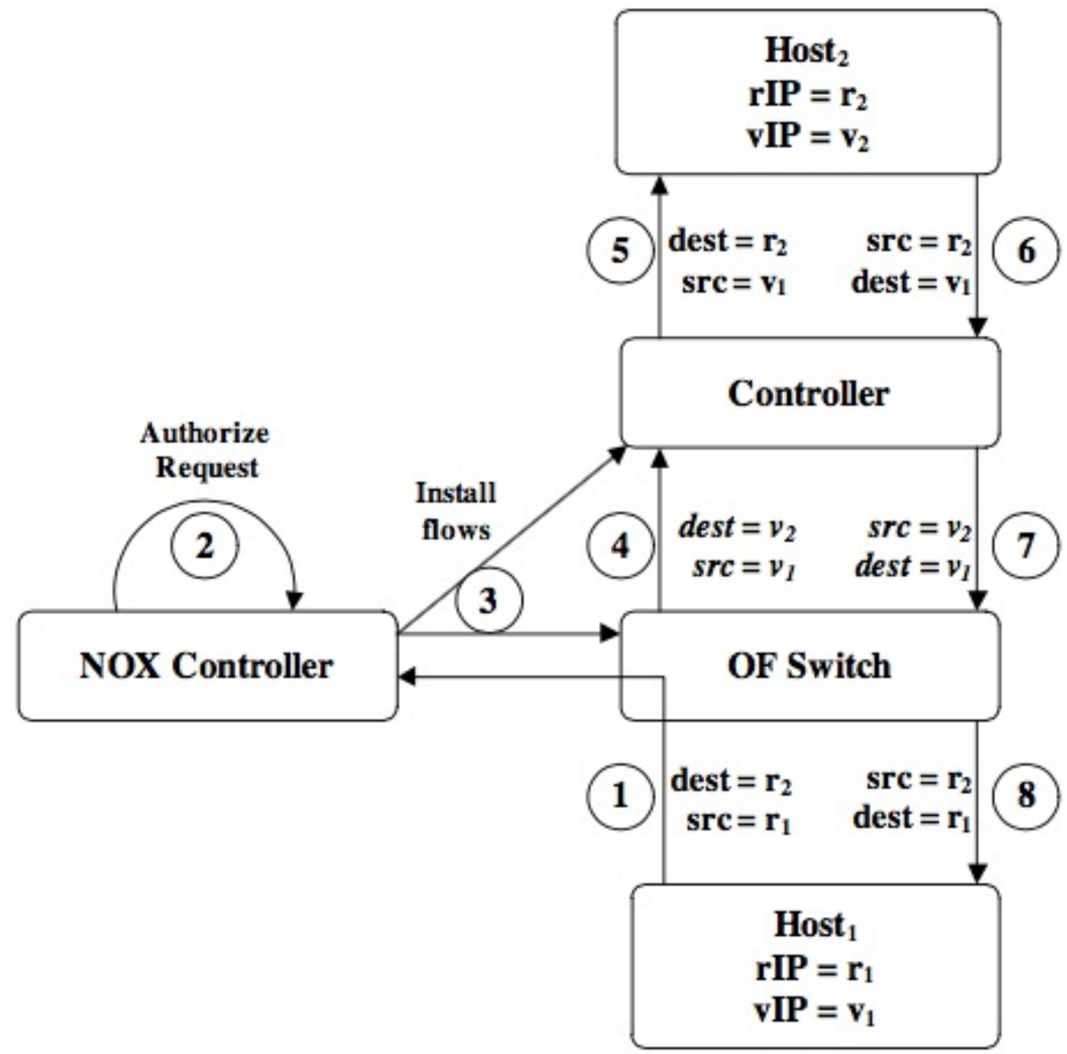


Figure 2: Communication via name



ıre 3: Communication via rIP address

Jafarian, Jafar Haadi, Ehab Al-Shaer, and Qi Duan. "Openflow random host mutation: transparent moving target defense using software defined networking." *Proceedings of the first workshop on Hot topics in software defined networks*. ACM, 2012.

Random Route Mutation

- View the whole network as a directed graph G=(V, E), V represents for hosts, E represents for links
- Assume there is a flow, which source is S, destination is D, the goal of RRM
  - find a route between S and D that satisfies the following constraints for every interval of the flow duration:
    - **Capacity constraint**: the new route should not include those nodes or links that do not have the bandwidth requirement for the flow;
    - **Overlap constraint**: to increase unpredictability and achieve good load balancing, the new route should avoid those intermediate nodes that appear in recently used routes;
    - **QoS constraint**: the mutated routes should maintain the required quality, such as bounded delays or number of hops.
  - If there are more than one satisfying routes for the flow, the RRM procedure should choose one satisfying route randomly.

Duan, Qi, Ehab Al-Shaer, and Haadi Jafarian. "Efficient random route mutation considering flow and network constraints." *Communications and Network Security (CNS), 2013 IEEE Conference on*. IEEE, 2013.

Random Route Mutation

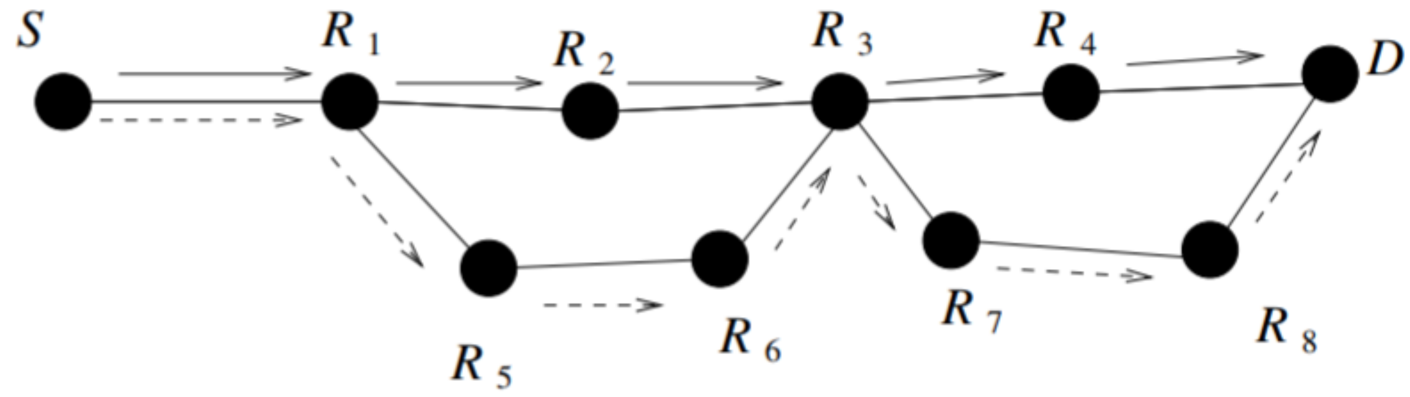


Fig. 1. An example of route change

- Overlay networks were proposed to improve the reliability of the Internet and to facilitate performance sensitive services,such as mission critical networks, high performance cloud computing etc., that are difficult or impossible to deploy natively on the Internet.
- We need to find a mapping between overlay nodes and substrate nodes with the following constraints:
  - **Unique Mapping Constraint**: every overlay node should be mapped to exactly one substrate node, and two distinct overlay nodes should not be mapped to the same substrate node.
  - **Disjoint Route Constraint**: the placement must guarantee that there are enough number of disjoint routes between the corresponding substrate source and destination of the overlay flows.
  - **Unpredictability Constraint**: there should be enough difference between the old and new placement.

问题和讨论